

NGUYEN LE QUOC BAO

AI Engineer · LLM Agents · Computer Vision · Generative Media

Vietnam · +84 326 032 296 · ngbao3558@gmail.com
GitHub: github.com/baonguyen3568 · Portfolio: baonguyen3568.github.io/myprofile

SUMMARY

AI Engineer with 14 months of production experience building LLM agents, real-time computer vision and generative media systems on self-managed NVIDIA infrastructure. Backend lead and release owner of an AI sales platform that shipped 14 releases in three weeks; sole engineer of a face-recognition attendance system running in production since April 2026 across two sites. Works across the full path from GPU bring-up and local model serving to agent design, product decisions and customer demos.

EXPERIENCE

AI Engineer & Product Manager — Nhat Tien Chung IT Telecoms (NTC) — Sep 2025 – Present

AI solutions team serving NTC and its affiliated companies (Thế Giới Máy Chủ, NTC AI). Lead project teams of up to 5 people; own architecture, task allocation and releases.

NTCsale — AI sales assistant & storefront platform · multi-tenant SaaS — Aug 2026 – Present
saleai.ntc.ai.vn

- Led backend development in a 5-person team (2 backend, 1 frontend, 2 QA) and owned every production release: **14 versions in 3 weeks**. Built the backend foundation, authentication and AI website generation on FastAPI, PostgreSQL, Redis, MinIO and Next.js.
- Refactored the codebase twice into a **modular monolith with ports-and-adapters layering**, enforced by an automated architecture checker; set the team's configuration model (DB/admin, environment, versioned prompt store).
- Designed the team's **multi-agent development workflow**: 12 role-specific coding agents (research, security, testing, fixing, design) sharing rules, skills and memory, coordinating through GitHub issues via MCP and escalating architecture decisions to me. The team merged **686 PRs** and closed **844 issues** in about four weeks.
- Simplified the sales assistant from a planner–executor design to a **single tool-calling loop** after the planner proved too unreliable for a self-hosted 27B model. Reorganised context into tiered memory and returned structured tool errors to the model so it recovers instead of falling back to canned replies.
- Narrowed the website builder's LLM to content and colour palettes over fixed templates, splitting truncated batches and degrading per slot to template text, with a bounded worst-case token cost per run.
- Resolved a production outage (a 5,762-byte CSP header exceeding Nginx's 4,096-byte limit): rolled back in under 17 minutes, shipped the fix in 27, and added a build-time guard so oversized policies fail the build instead of the deploy.
- In pilot with an affiliated company: website chat and product management across about 1,600 products.

NTC VisionID — Face-recognition attendance · real-time edge AI — Apr 2026 – Present
vision.ntc.ai.vn

- **Sole engineer from design to production** (second maintainer onboarded Oct 2026): 3 affiliated companies, **9 RTSP cameras** across Ho Chi Minh City and Hanoi, about **60 employees** checked in daily.
- Built the GStreamer → SCRFD detection → ArcFace embedding → tracking pipeline, sustaining **15–20 FPS per camera** with 20–30 people in frame on a single RTX PRO 2000 Blackwell; migrated it across Jetson, RTX PRO 5000 and RTX PRO 2000.
- Tuned the cosine-similarity threshold from 0.70 to 0.55 using operational feedback, removing slow check-ins with no reported misidentification since; replaced delayed email/Telegram alerts with a PWA using web push.
- Delivered the HR layer on top: shift rules, leave requests, attendance explanations, overtime, multi-level approvals, role-based access and payroll export.

blog2vid — Fully local AI video production · in progress — Oct 2026 – Present

- Leading a 7-agent build of an article-to-video pipeline on a DGX Spark: Qwen3.8-27B script → Qwen-Image-Edit stills → **MiniMax H3** video with native audio → VieNeu-TTS voice cloning → ACE-Step music → HyperFrames composition, on FastAPI, MongoDB, Redis and Kafka.
- Selected models on public benchmarks and licence review (dropped LTX over licensing; moved from Wan 2.2 to H3). Hardened the shared machine with OOM guards, a 50% memory cap and a thermal watchdog after hangs at 97 °C.

AI infrastructure, PoCs & pre-sales Sep 2025 – Present

- Set up and operate a heterogeneous NVIDIA fleet from bare OS: DGX Spark (GB10), RTX PRO 5000/2000 Blackwell servers, Jetson edge devices and Xeon Gold VPS. Serve local LLMs with vLLM (Qwen3.8-27B NVFP4, FP8 KV cache) for internal products and customer PoCs.
- Built the **document agent RAG** on labs.ntc.ai.vn, NTC's catalogue of apps that customers install on DGX Spark with one command: OCR and vision-model parsing, hybrid lexical + vector retrieval on OpenSearch, reranking, and a tool-calling agent that answers with file and page citations. Assigned the team to package NVIDIA blueprints for the same catalogue to support GB10 sales.
- Represented NTC at **DCCI Summit 2026** (Viettel IDC, HCMC — edge AI cameras on Jetson), **AI Day 2026** (Hanoi — NemoClaw/Hermes agent runtimes and agentic coding) and a **TD SYNNEX** partner event (HCMC — RAG assistant for banking): talks, booth demos and pre-sales consultations.
- Wrote technical content and proposals, including an article on NVIDIA Vera Rubin at OCP 2026 and an NVIDIA AI Factory Operations Agent proposal for enterprise customers.

TECHNICAL SKILLS

LLM & agents	Tool-calling agents, LangGraph, LangChain, context engineering, tiered memory, MCP, agentic RAG, hybrid retrieval (OpenSearch, Qdrant), reranking, Docling, structured outputs, OpenAI/Anthropic-compatible gateways, NemoClaw/Hermes runtimes, multi-agent coding workflows
Model serving	vLLM, NVFP4/FP8 quantised inference, DGX Spark (GB10), RTX PRO Blackwell, Jetson, CUDA, GPU memory and thermal tuning
Computer vision	InsightFace (SCRFD, ArcFace), GStreamer, RTSP, multi-camera tracking, vector search, threshold tuning
Generative media	MiniMax H3, Wan 2.2, Qwen-Image-Edit, ComfyUI, VieNeu-TTS voice cloning, ACE-Step, NVIDIA Riva
Backend & platform	Python, FastAPI, SQLAlchemy/asyncpg, PostgreSQL, MongoDB, Redis, Kafka, MinIO/S3, Nginx, Docker Compose, TypeScript, Next.js, React
Engineering practice	Modular monolith / ports and adapters, architecture decision records, CI/CD and release management, incident response, code review

EDUCATION

Engineer's Degree in Artificial Intelligence — Nguyen Tat Thanh University 2021 – 2025
GPA 3.32 / 4.00 · Classification: Very Good

LANGUAGES

Vietnamese (native) · English (professional technical reading)